Quyen Tran

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Research Interests

Visualization, Human–Computer Interaction (HCI), AI in Education (AIED), with a focus on designing visual and interactive systems that make abstract concepts and formal reasoning more understandable and accessible.

Education

Aug 2023 to May 2027	**B.A. in Computer Science and Mathematics**, GPA: 3.95/4.0, majors: 4.0/4.0 *DePauw University, Greencastle, IN*
Aug 2025 to Dec 2025	**Budapest Semester in Mathematics**, GPA: 4.0+ /4.0 Faculty affiliation: *Alfréd Renyi Institute of Mathematics & Eötvös Loránd University, Hungary*

Publications

[3] **Quyen Tran** and Jyotirmoy Sarkar. "Ranking Events Based on a Random Sample." *Resonance—Journal of Science Education*, Classroom Section, 2026. [Paper]

[2] Jyotirmoy Sarkar and **Quyen Tran**. "Predicting Outcomes of Random Phenomena." *International Journal of Mathematics, Statistics and Operations Research*, 5(2), 2025, pp. 287–302. [Paper]

[1] **Quyen Tran** and Mamunur Rashid. "Optimizing Transport Predictive Modeling with Simulation-Based Statistical Inference." *International Journal of Statistical Sciences*, 24(20), Dec. 2024, pp. 163–179. [Paper]

Preprints & Manuscripts

[1] **Quyen Tran**, Hieu Tran, Sutthirut Charoenphon, and Brian T. Howard. "CUMath: A Benchmark for Evaluating LLM Step-by-Step Reasoning in Undergraduate Mathematics" *Preprint*, 2026. [Paper]

Presentations & Talks (* denotes equal contribution)

[3] **Quyen Tran**, Hwei-Shin Harriman, Dominik Moritz, and Joshua Sunshine. "Automating Textbook Proof Conversion and Dataset Generation for Geometry Learning." Poster presented at the *REUSE Poster Session*, Carnegie Mellon University, Pittsburgh, PA, 2026. [Poster]

[2] Vasudev Menon*, Sam Reiter*, Jayson Tran*, **Quyen Tran***, Benjamin Whitsett*, and Pál Zsámboki. "Mechanistic Interpretability and Simple Games." Research talk presented at the *Budapest Semesters in Mathematics Undergraduate Research Program*, Budapest, Hungary, 2025. [Slide]

[1] **Quyen Tran** and Mamunur Rashid. "Optimizing Transport Predictive Modeling with Simulation-Based Statistical Inference." Poster presented at *MAA MathFest*, Indianapolis, IN, Aug. 2024. [Poster]

Selected Technical Papers

[3] **Quyen Tran**. "Learning Nim Strategies with Bayesian Reinforcement Learning and MCMC." Final paper, *Math Senior Seminar, DePauw University*, 2026. [PDF]

[2] **Quyen Tran** and Hung Nguyen. "Clustering-Based Lightweight Architecture for Image Captioning." Final project paper, *CSC 370: Data Mining, DePauw University*, 2026. [PDF]

[1] **Quyen Tran**. "Sketch- and Text-Guided Image Generation with Diffusion Models." Final project paper, *Introduction to Deep Learning, Budapest Semesters in Mathematics*, 2025. [PDF]

Research Experience

May 2026 to Present

Research Intern (REUSE Program) — **Carnegie Mellon University**
Software and Societal Systems Department

Project: Automating Geometric Proof (Poster [3])
Mentors: Hwei-Shin Harriman, Dr. Dominik Moritz, Dr. Joshua Sunshine

- Working on Ender, an interactive system that visualizes high school geometric proof states and dependencies to support students' proof understanding and error identification.
- Developed a pipeline that successfully converted over 70% of selected textbook geometry proofs from images and PDFs into Ender's formal syntax.
- Designed constraint-aware proof mutation methods using theorem-relation and geometric-object graphs, generating 696 synthetic incorrect proofs and identifying 11 proof-checker bugs.
- Formalizing Ender's proof language and correctness in Rocq; manuscript in preparation for submission to **OOPSLA** or **PLDI** 2027.

Jan 2026 to Present

Research Intern — **SEACrowd**
Reasoning Agentic LLM Router Research Group
Mentors: David Anugraha (MS @ Stanford), Dr. Genta Winata (Scientist @ Capital One)

- Developing reasoning-based LLM routing strategies for selecting the most suitable model per query, with a focus on routing performance and failure cases.
- Evaluating routing across 20 LLMs on 15 benchmarks across mathematics, coding, and knowledge using LLMRouterBench, and investigating limitations of existing evaluation methods.

Sep 2025 to Dec 2025

Student Researcher — **Budapest Semester in Mathematics**
Project: Mechanistic Interpretability & Simple Game (Slides [2])
Advisor: Dr. Pál Zsámboki (HUN-REN Rényi Institute, AI Group)

- Conducted experiments on 10^2–10^4 SGD-trained single-layer transformers for structured (n, k) tic-tac-toe across varying embedding and key dimensions.
- Created visualizations to analyze learned embedding geometry, attention patterns, and failure modes, comparing gradient-learned models with theoretically constructed solutions.

May 2024 to Present

Undergraduate Researcher — **DePauw University**
Departments of Computer Science and Mathematical Sciences

Project 2: Evaluating LLM Step-by-Step Reasoning in Undergraduate Mathematics (Preprint [1])
Advisors: Hieu Tran (PhD @ Purdue), Dr. Sutthirut Charoenphon, Dr. Brian Howard

- Conducted an IRB-approved campus-wide study of undergraduate students' use of and trust in generative AI for coursework.
- Developed CUMath, a human-annotated benchmark of 2,100 undergraduate math problems with expert-written step-by-step solutions, and evaluated 15 LLMs across 4 prompting strategies.
- Built automated and LLM-as-a-judge evaluation pipelines using mathematical reasoning metrics, MathBERT representations, and Wolfram Alpha verification.
- Conducted human evaluation and quantified agreement among automated, LLM-based, and independent human judgments.

Project 1: Optimizing Transport Predictive Modeling with Simulation-Based Statistical Inference (Publication [1], Poster [1])
Advisor: Dr. Mamunur Rashid

- Investigated simulation-based inference methods for transportation prediction models, applying ABC–MCMC and Synthetic Likelihood to train GBM models under limited-data settings.
- Evaluated predictive performance across simulated and observed transportation data, achieving prediction errors below 10%.

Independent Projects

Jan 2026 to May 2026 — **Project: Bayesian Reinforcement Learning for the n-Pile Nim Game** (Paper [3])
Math Senior Seminar Project, DePauw University

Explored how Bayesian reinforcement learning learns interpretable game-playing strategies in the n-pile Nim game using Bayesian policy learning and MCMC-based inference.

Jan 2026 to May 2026 — **Project: Clustering-Based Lightweight Architecture for Image Captioning** (Paper [2])
Data Mining Course Project, DePauw University

Investigated clustering-based methods for improving the specificity and contextual relevance of image captions across ViT-GPT2, GIT, and BLIP models.

Sep 2025 to Dec 2025 — **Project: Sketch-Guided Diffusion for Image-to-Image Generation** (Paper [1])
Deep Learning Course Project, Budapest Semesters in Mathematics

Studied sketch-guided diffusion models for controllable image generation, with an aim to better align generated images with human structure and intent.

Mar 2025 to Oct 2025 — **Project: Explainable Breast Cancer Detection Model**
Advisor: Dr. Mehmet Gulum (DePauw University)
Co-authors: Hieu Tran (PhD @ Purdue), Tri Dang (PhD @ UC Merced), Dat Nguyen (MS @ Syracuse)

Built a Vision Transformer for breast cancer prediction with attention-based explanations to help clinicians interpret and verify model predictions.

Aug 2024 to Apr 2025 — **Project: Statistical Structure of Random Event Sequences** (Publications [2, 3])
Advisor: Dr. Jyotirmoy Sarkar (Indiana University–Indianapolis)

Studied finite-sample statistical structure in random phenomena for ranking and prediction under probabilistic models.

Teaching Assistant & Services

Teaching

Fall 2026	STEM Guide (Teaching Assistant & Tutor) for CSC 121: Introduction to Computer Science
Spring 2026	Teaching Assistant for CSC 121: Introduction to Computer Science
Spring 2025	Lab Assistant for CSC 121: Introduction to Computer Science
2024 – 2025	Math Tutor: Discrete Math, Intro to Proof, Calculus I–III, Linear Algebra
Spring 2024	Math Tutor: Discrete Math, Calculus I–II

Services

2023 – 2027	Vice President at DePauw Math Club
2024 – 2025	Technical Advisor at DePauw Women in Computer Science

Honors, Awards, and Grants

2023 – 2027	Science Research Fellow, DePauw University
2023 – 2027	Dean's List (every semester), DePauw University
2026	REUSE Internship Scholarship, Carnegie Mellon University
2026	Computer Science Honor Society, DePauw University
2025	Waldemar J. Trjitzinsky Memorial Award, American Mathematical Society
2025	Kitüntetés (academic distinction), Budapest Semester in Mathematics
2025	Charles and Frances (Wylie) Condit Science Math Award, DePauw University

2025	J. William Asher and Melanie J. Norton Fund in the Sciences, DePauw University
2024	Student Research and Artistic Grants, DePauw University
2024	Faculty Development Committee Summer Research Funding, DePauw University
2024	Second Place, Indiana Collegiate Mathematics Competition (ICMC)
2023	Distinguished Scholarship (2023–2027), DePauw University

Technical Skills

Languages: Python, Java, C++, R, JavaScript, HTML/CSS, SQL

Libraries: PyTorch, Hugging Face, scikit-learn, fastai, TensorFlow, OpenCV, NumPy, pandas, matplotlib

Tools: LaTeX, Visual Studio Code, Google Colab, Git/GitHub